

7.5 (a) We will use the result from Problem 7.9. Here, for the flat sheet, we can write

$$k = \frac{1+i}{\delta} = \frac{n\omega}{c}, \text{ or equivalently, } n = \frac{(1+i)c}{\omega\delta}, \text{ and } \alpha_1 = n\omega D/c = (1+i)D/\delta. \text{ We can also}$$

set $\alpha_2 = 0$. Then,

$$T = \frac{1}{4n} \begin{bmatrix} 4n \cos \alpha_1 I + i \sigma_3 2(n^2+1) \sin \alpha_1 & -\sigma_2 2(n^2-1) \sin \alpha_1 \\ 4n \cos \alpha_1 + 2i(n^2+1) \sin \alpha_1 & -2i(n^2-1) \sin \alpha_1 \\ -2i(n^2-1) \sin \alpha_1 & 4n \cos \alpha_1 - 2i(n^2+1) \sin \alpha_1 \end{bmatrix}$$

Using 7.8 (c),

$$\frac{E_r}{E_i} = -\frac{t_{21}}{t_{22}} = \frac{2i(n^2-1) \sin \alpha_1}{4n \cos \alpha_1 - 2i(n^2+1) \sin \alpha_1}$$

For a good conductor, $|n| \gg 1$, then

$$\frac{E_r}{E_i} \rightarrow \frac{e^{i\alpha_1} - e^{-i\alpha_1}}{\frac{2}{n}(e^{i\alpha_1} + e^{-i\alpha_1}) - (e^{i\alpha_1} - e^{-i\alpha_1})} = \frac{-(1 - e^{2i\alpha_1})}{(1 - e^{2i\alpha_1}) + \frac{2}{n}(1 + e^{2i\alpha_1})}$$

Since $i\alpha_1 = -\frac{(1-i)D}{\delta}$, $\frac{2}{n} = \frac{2}{1+i} \frac{\omega\delta}{c} = (1-i) \frac{\omega\delta}{c}$, then

$$\frac{E_r}{E_i} = \frac{-(1 - e^{-2\lambda})}{(1 - e^{-2\lambda}) + \gamma(1 + e^{-2\lambda})}, \text{ with } \lambda = (1-i)D/\delta \text{ and } \gamma = (1-i) \frac{\omega\delta}{c} = \frac{2}{n}$$

$$\text{Similarly, } \frac{E_t}{E_i} = \frac{\det T}{t_{22}} = \frac{\frac{1}{16n^2} (16n^2 \cos^2 \alpha_1 + 4(n^2+1)^2 \sin^2 \alpha_1 - 4(n^2-1)^2 \sin^2 \alpha_1)}{\frac{1}{4n} (4n \cos \alpha_1 - 2i(n^2+1) \sin \alpha_1)}$$

$$= \frac{4n}{4n \cos \alpha_1 - 2i(n^2+1) \sin \alpha_1} = \frac{4/n}{\frac{2}{n}(e^{i\alpha_1} + e^{-i\alpha_1}) - (e^{i\alpha_1} - e^{-i\alpha_1})}$$

$$= \frac{2\gamma e^{i\alpha_1}}{\gamma(e^{2i\alpha_1} + 1) - (e^{2i\alpha_1} - 1)} = \frac{2\gamma e^{-\lambda}}{(1 - e^{-2\lambda}) + \gamma(1 + e^{-2\lambda})}$$

(b) For $D \rightarrow \infty$, $\gamma \rightarrow \infty$, $E_r \rightarrow 0$, $E_t \rightarrow 0$. No reflection and no transmission. As $D \rightarrow \infty$, there is no transmission as all the energy has been dissipated.

For $D \rightarrow 0$, $\gamma \rightarrow 0$, $E_r = 0$, $E_t = 1$. Propagation in air, as if nothing happens

$$(c) \quad T = \left| \frac{E_t}{E_i} \right|^2 = \left| \frac{2\gamma e^{-D/\delta} e^{iD/\delta}}{(1 - e^{-2D/\delta} e^{2iD/\delta}) + \gamma(1 + e^{-2D/\delta} e^{2iD/\delta})} \right|^2$$

$$\text{As } \left| (1 - e^{-2D/\delta} e^{2iD/\delta}) + \gamma(1 + e^{-2D/\delta} e^{2iD/\delta}) \right|^2$$

$$\approx 1 - 2e^{-2D/\delta} \cos(2D/\delta) + e^{-4D/\delta}$$

and $|\gamma|^2 = 2(\text{Re } \gamma)^2$, then

$$T = \frac{8(\text{Re } \gamma)^2 e^{-2D/\delta}}{1 - 2e^{-2D/\delta} \cos(2D/\delta) + e^{-4D/\delta}}$$